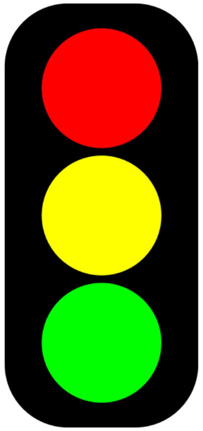
7.9.6 Wrap-Up: Reflection

1. Circle the color on the stoplight that best represents your understanding of solving real-world problems using an angle of depression or an angle of elevation.



Red: I do not understand.

Yellow: I need some more help.

Green: I got this; I understand!

2. Complete the following statements if you selected red or yellow.

   a. I am still unsure about …

   b. To fill in this gap I intend to …

Unit 7, Lesson H1: Law of Sines

Benchmark

MA.912.T.1.3 Apply the Law of Sines and the Law of Cosines to solve mathematical and real-world problems involving triangles.

Learning Target

☐ I can solve mathematical and real-world problems involving triangles using the Law of Sines.

7.H1.1 Warm-Up: Notice and Wonder

1. A theodolite is an instrument used for measuring angles between designated visible points on the horizontal and vertical planes. Theodolites are often used for land surveying.

Late 16th century Theodolite	Modern Day Theodolite
	

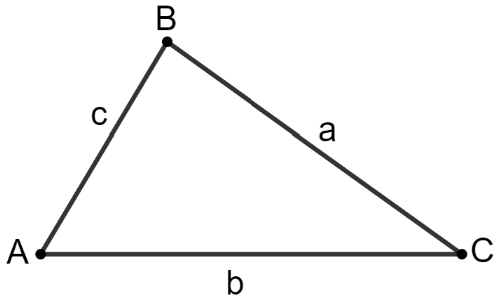
a. What do you notice?

b. What do you wonder?

7.H1.2 Collaboration: Deriving the Formula for Law of Sines

Right triangle trigonometry can be used to solve problems involving right triangles, but not all triangles are right triangles. If a triangle is not a right triangle, the triangle's altitude can be used to create a right triangle so that right triangle trigonometry can be used.

1. ΔABC is shown, where $AB = c$, $AC = b$, and $BC = a$.



a. Use ΔABC to do the following.

- Sketch an altitude from vertex B.
- Label the altitude k.

The altitude creates two right triangles inside ΔABC. Notice that $\angle A$ is contained in one of the right triangles, and $\angle C$ is contained in the other.

b. Complete each equation using right triangle trigonometry.

$\sin(A) =$ $\sin(C) =$

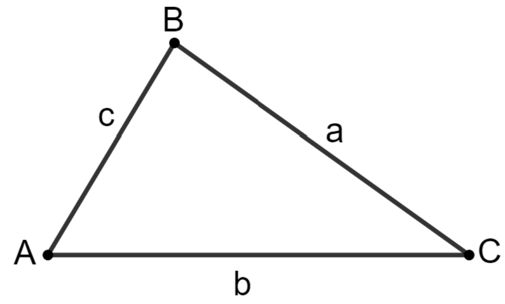
c. Discuss with your partner why both equations use k. Summarize your discussion.

d. Solve each equation for k.

e. Set the equations equal to each other to form a new equation.

f. Rewrite the new equation by regrouping a with $\sin(A)$ and c with $\sin(C)$.

2. ΔABC is shown where $AB = c$, $AC = b$, and $BC = a$.



a. Use ΔABC to do the following.

- Sketch an altitude from vertex C.
- Label the altitude k.

The altitude creates two right triangles inside ΔABC. Notice that $\angle A$ is contained in one of the right triangles, and $\angle B$ is contained in the other.

b. Complete each equation using right triangle trigonometry.

$sin\,(A) =$ $\qquad$ $sin\,(B) =$

c. Solve each equation for k.

d. Set the equations equal to each other to form a new equation.

e. Re-write the new equation by regrouping a with $sin\,(A)$ and b with $sin\,(B)$.

The **Law of Sines** is $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

The Law of Sines may also be written as

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

7.H1.3 Guided Instruction: Law of Sines

1. Some of the measurements of ΔDLC are $DL = 17$, $m\angle DLC = 40°$, and $m\angle CDL = 75°$.Determine the measurement of each part of ΔDLC. Round to the nearest tenth.

Part	Measurement
$\overline{LC}$	
$\overline{DC}$	
$\angle DCL$	

2. In ΔRFM, $RM = 66,\ FM = 30$, and $m\angle RFM = 26°$. Determine the measurement of each part of ΔRFM. Round to the nearest tenth.

Part	Measurement
$\overline{FR}$	
$\angle FRM$	
$\angle FMR$	

7.H1.4 Guided Practice: Law of Sines

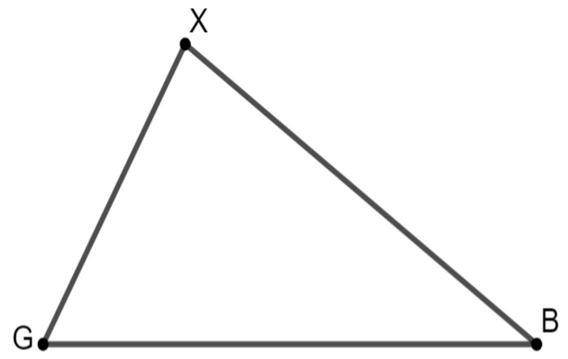
1. ΔLVY has side measurements of $VY = 8.5$ and $LY = 16.5$. The measure of $\angle LVY$ is $84°$. Find all missing sides and angles. Round to the nearest tenth.

2. The three known measurements of ΔDNH are $DN = 26$, $m\angle NDH = 80°$, and $m\angle DNH = 75°$. Find all missing sides and angles. Round to the nearest tenth.

7.H1.5 Your Turn!

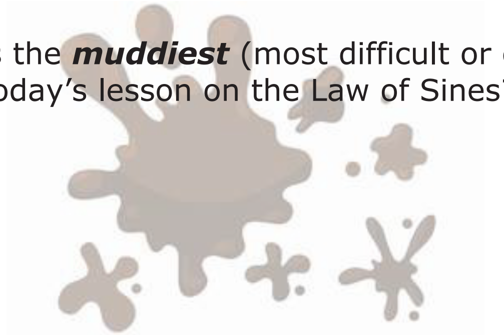
1. In ΔLSK, $LS = 7$, $m\angle KLS = 31°$, and $m\angle KSL = 107°$. Determine the length of $\overline{KS}$. Round to the nearest tenth.

2. Cell phone towers represented by points G and B are 25 miles apart. A new tower, X, is being built so that $m\angle G = 54°$ and $m\angle B = 26°$. How far from tower G will tower X be built? Round to the nearest tenth.



7.H1.6 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson on the Law of Sines?



Unit 7, Lesson H2: Law of Cosines

Benchmark

MA.912.T.1.3 Apply the Law of Sines and the Law of Cosines to solve mathematical and real-world problems involving triangles.

Learning Target

☐ I can solve mathematical and real-world problems that involve triangles using the Law of Cosines.